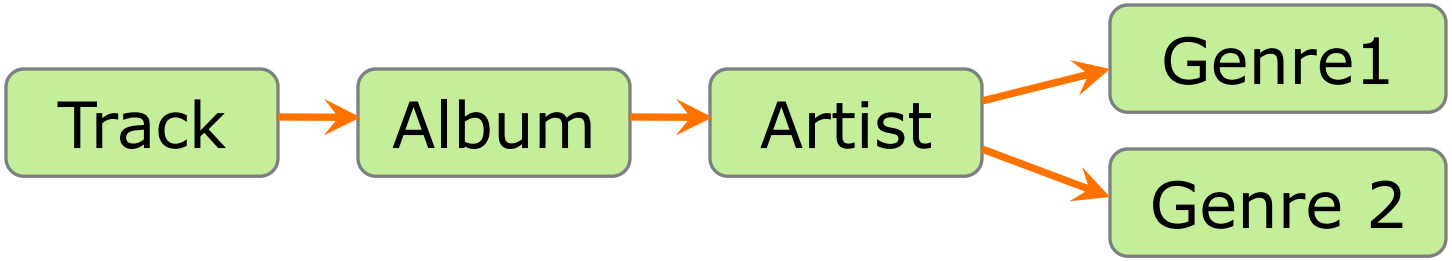
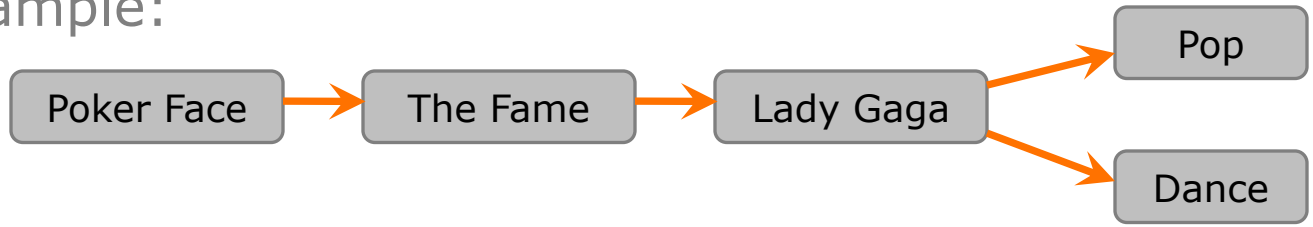
Music Recommendations

Yahoo! Music Dataset

- Music Data is stored in hierarchy structure:
  - A song track is in an album
  - An album is with an artist
  - An artist is with one or more genres
  - User can rate any item with score between 0~100.



- Example:



Track 2: Training/Test Data Format

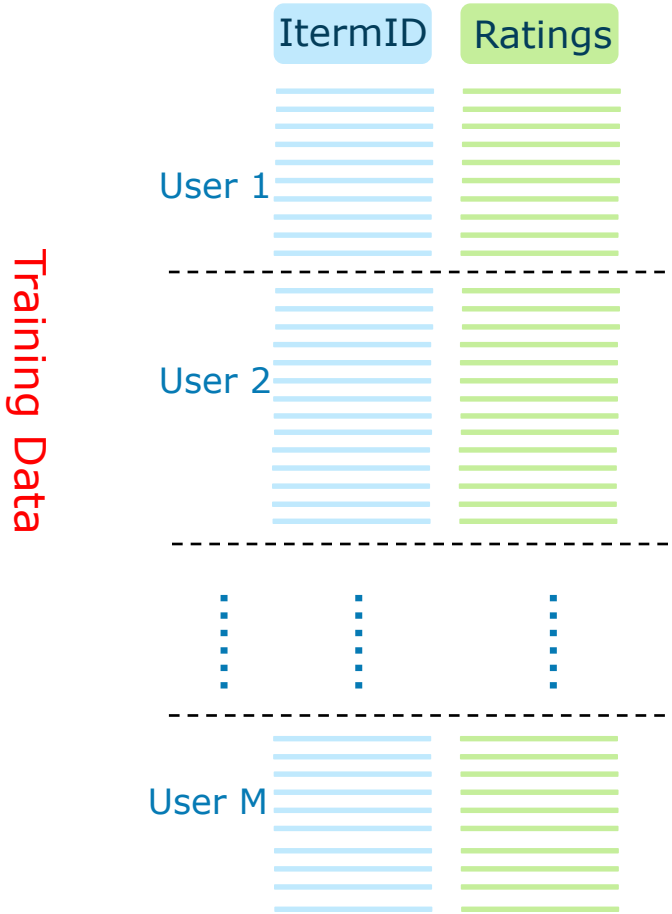
Training Data

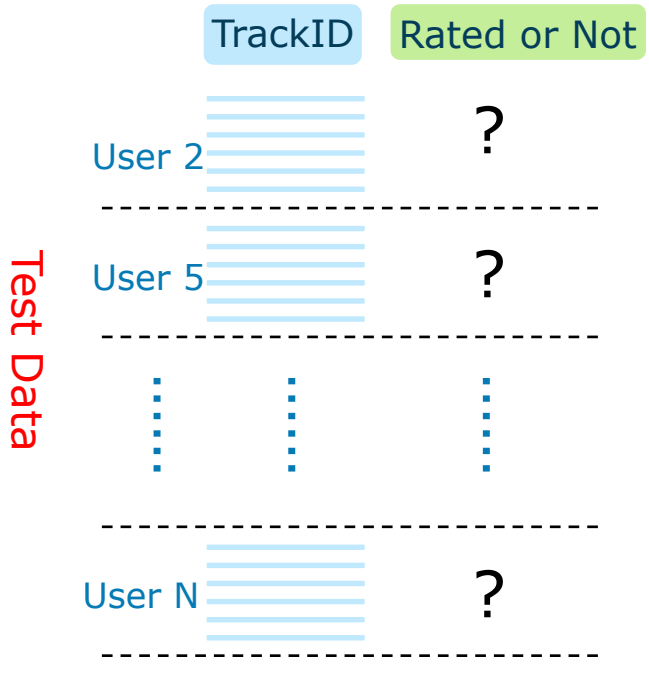
	ItermID	Ratings
User 1		
User 2		
⋮	⋮	⋮
User M		

Hierarchy List

Track → Album → Artist → Genre 1 ··· Genre K

Track 2: Training/Test Data Format





Hierarchy List

Track → Album → Artist → Genre 1 ▪▪▪ Genre K

Track 2: Data Statistics

- Training Data:
  - 249K users
  - 296K items
  - 62M ratings scores
- Test Data:
  - 100K test users, 6 tracks per user
  - Test user in the training user set

	track%	album%	artist%	genre%
Training data	43.6%	31.2%	19.1%	6.0%
Test data	100%			

Data Snapshot for Single User

Test Data

Item	Rated or Not
Track ID 1	?
Track ID 2	?
Track ID 3	?
Track ID 4	?
Track ID 5	?
Track ID 6	?

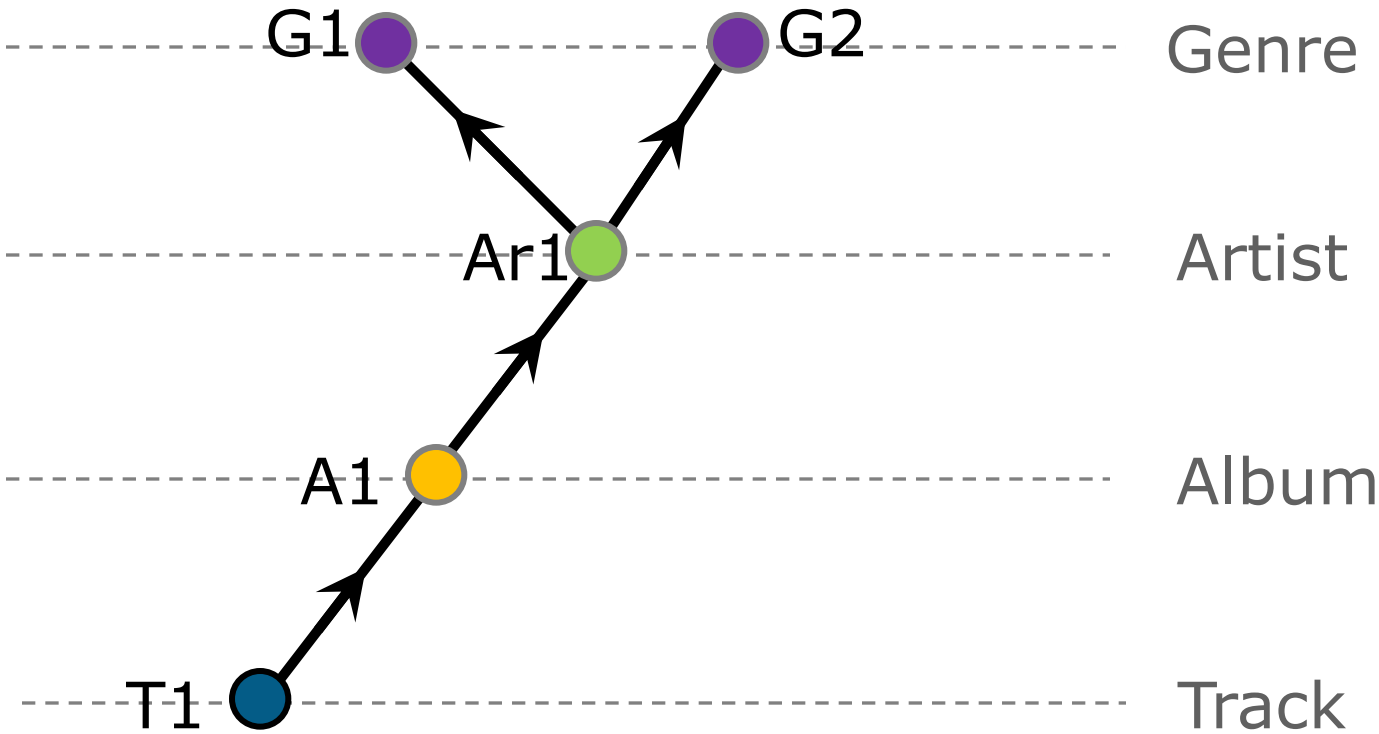
Task:

Find 3 tracks rated by the user from 6 high-rated tracks, i.e., determine 3 "1" and 3 "0".

Training Data

Item	Rating
Track ID	90
⋮	⋮
Track ID	40
Album ID	100
⋮	⋮
Album ID	80
Artist ID	90
⋮	⋮
Artist ID	100
Genre ID	0
⋮	⋮
Genre ID	70

Hierarchy Search



Example for User 249008

	TrackID		Album	Artist	Genre1	Genre2	Genre3	Genre4	Genre5	Genre6
Track 1	4967	Hierarchy List	205719	197877	172023	201738				
Track 2	164591	Hierarchy List	137283	51948	33722	131552	192353	142021	224280	12274
Track 3	127497	Hierarchy List	245158	218424	33204	239725				
Track 4	165413	Hierarchy List	116255	276506	61215	17453	212410	201738	88853	
Track 5	197975	Hierarchy List	119180	211565	33722	271945	224280			
Track 6	239621	Hierarchy List	262661	134540	61215	212410				

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		History Rating	80	90		80				
Track 3	127497	Hierarchy List	245158	218424	33204	239725				
		History Rating	50							
Track 4	165413	Hierarchy List	116255	276506	61215	17453	212410	201738	88853	
		History Rating	80	90		80				
Track 5	197975	Hierarchy List	119180	211565	33722	271945	224280			
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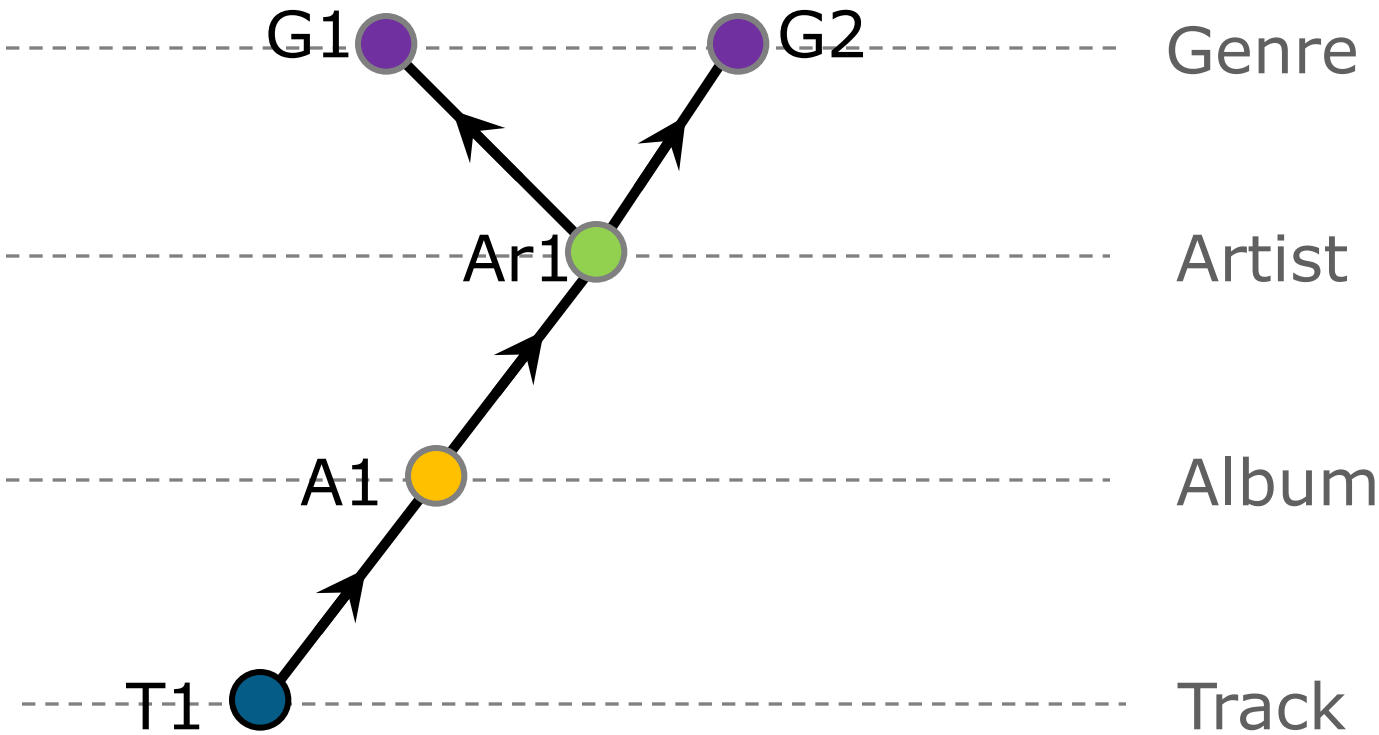
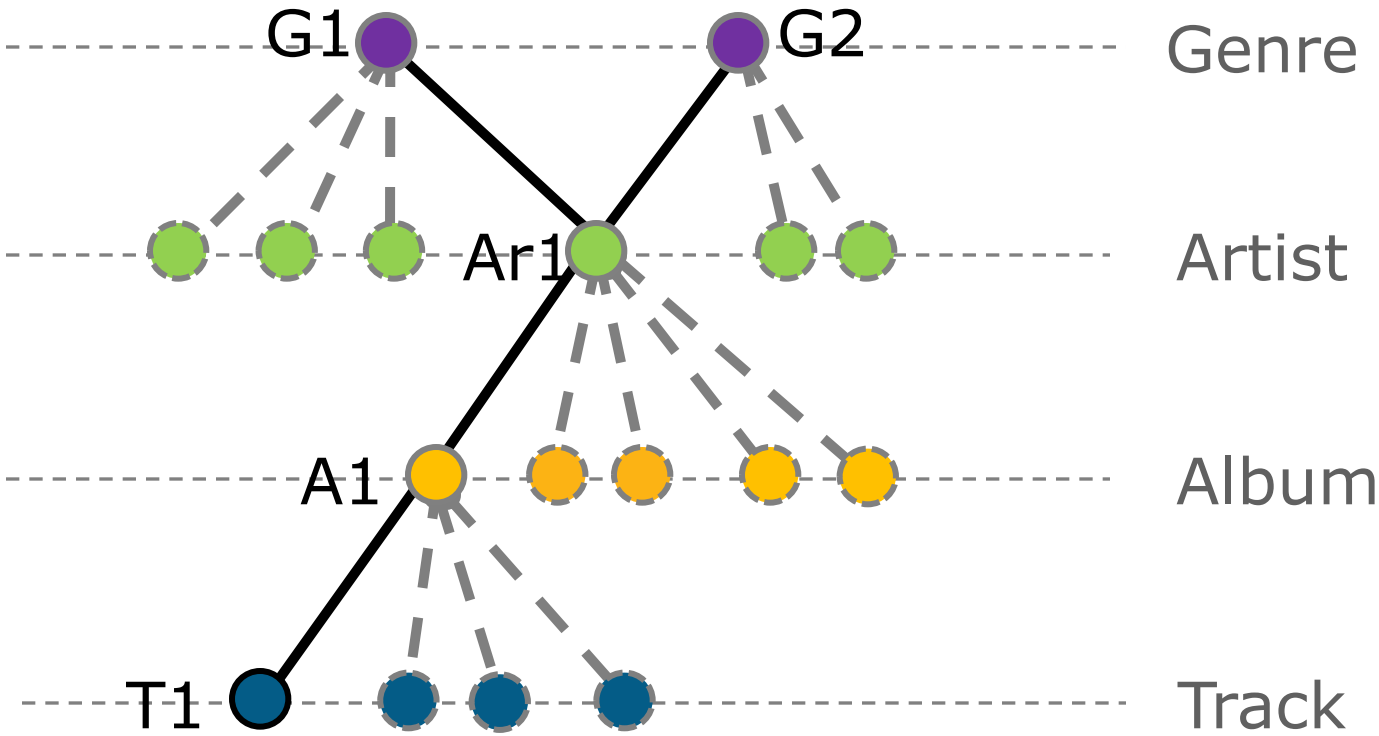
Prediction from Rating Scores

1. Simple rule-based method can be applied.
   - 13% error rate (87% accuracy achieved)
2. Generate features from the historic rating scores
   - Validation set needed
   - Album and Artist have only one rating score
   - Genres have more than one score.
     Statistical measurements
     - Maximum/Minimum Score
     - Average Score
     - Variance
     - Score Length
   - 10% error rate

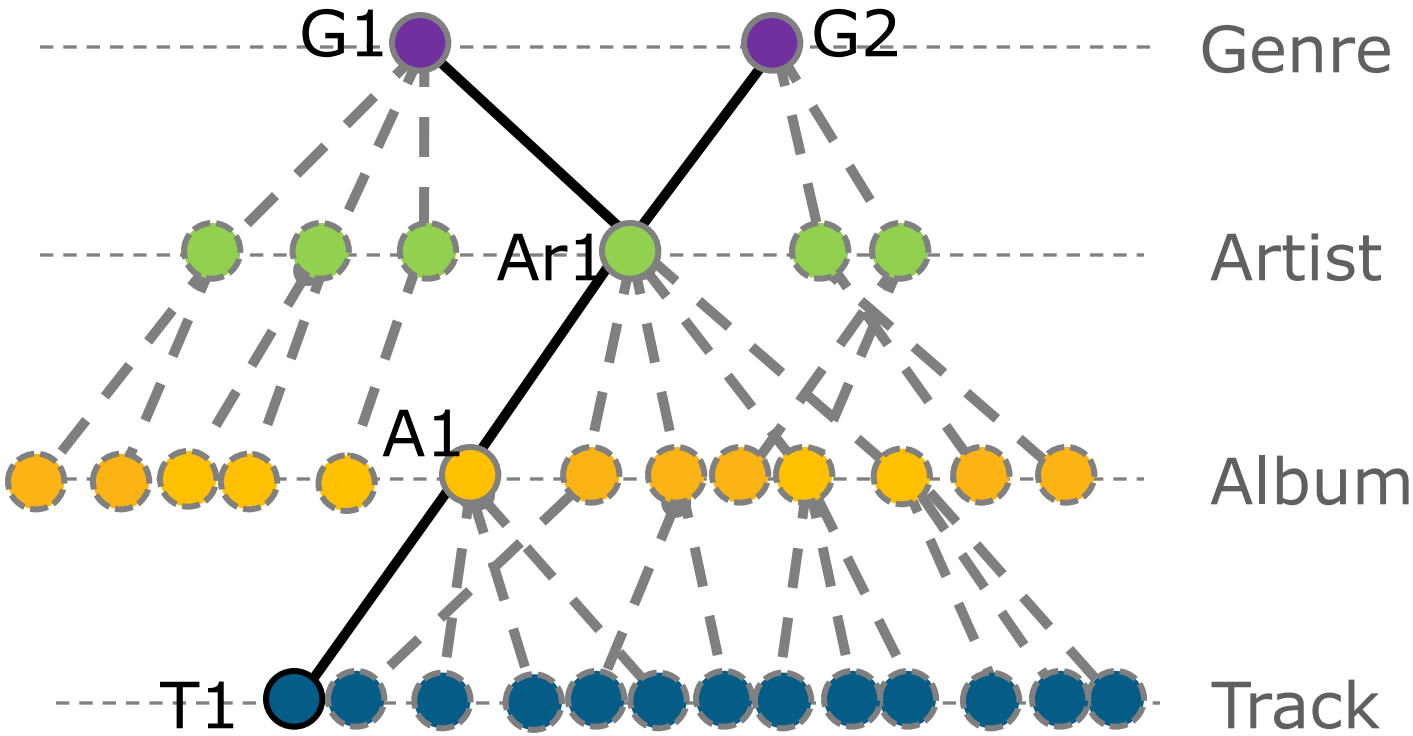
Hierarchy Search



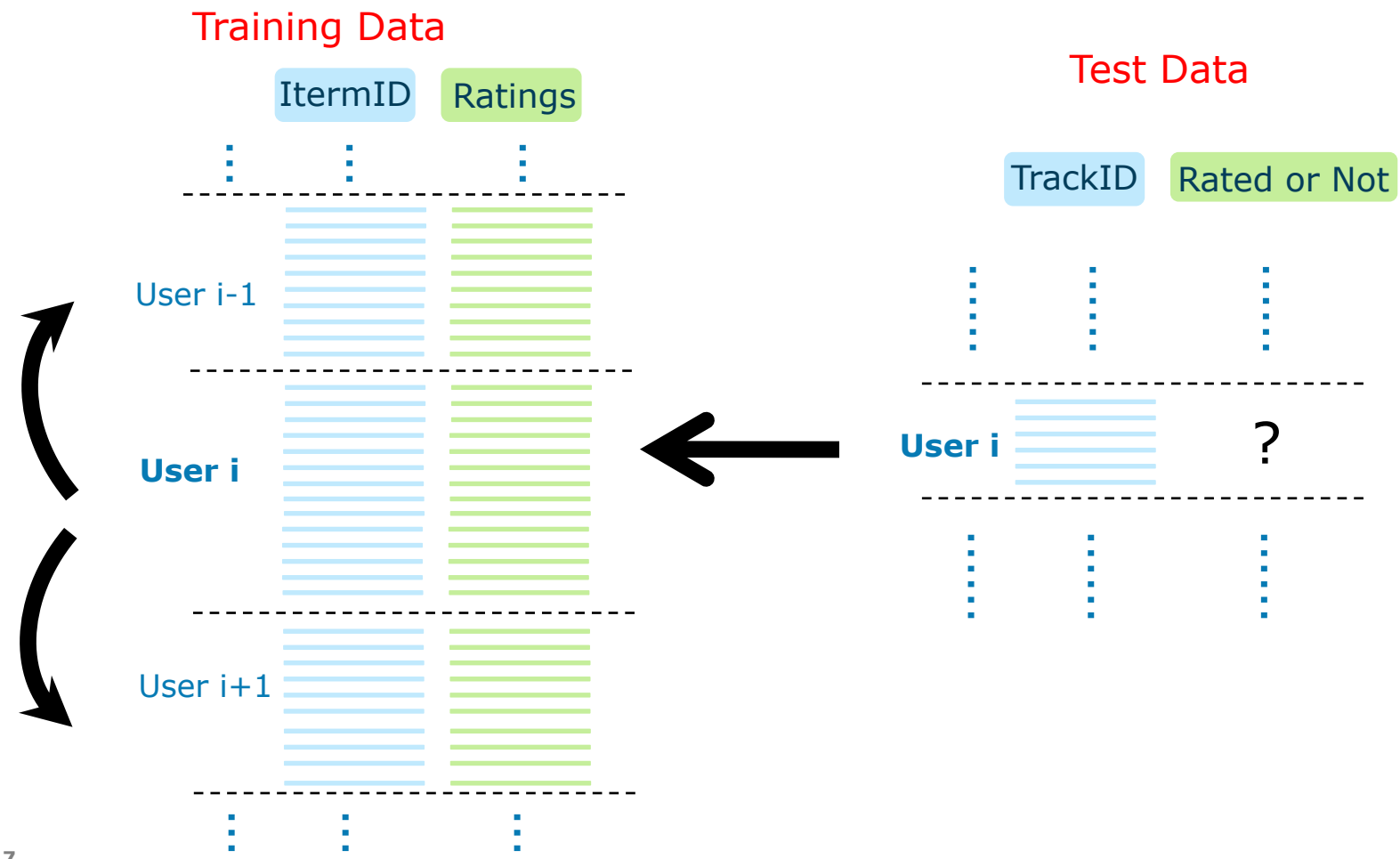
Expansion of Rating Hierarchy



Expanding More



More Features from Similar Users

- So far is user-driven prediction
- Non-test user information is ignored
- For users with little set of rating records
  - Hierarchy search and expansion end with none
  - Most errors come from these users with sparse information
- Expand search to all users in the training set
  - Match similar behavior users with larger set of records
  - Borrow features from similar users

Similar Users Match